

one framework, three climate twins

	wildfire	flood	urban heat and infrastructure
Perceive	satellite fire mask, weather, fuel and terrain give the fire front field	radar water extent, gauges and rainfall nowcast give the water depth field	satellite surface temperature and grid load give the temperature field
Simulate	next day fire spread with the hybrid FNO and U-Net model	shallow water routing with an FNO surrogate	heat and pollutant transport with an FNO surrogate
Plan	firebreaks, crews and evacuation within crew budget, crew safety and equity limits	gates, pumps and barriers within energy budget, hospital depth cap and equity limits	cooling centres and load shedding within energy budget, exposure cap and equity limits
Explain	incident action brief with a faithfulness certificate	operations brief for emergency managers	public advisory and utility dispatch note
PDE testbed	reaction diffusion (rdf), real NDWS data	shallow water equations (swe, PDEBench)	diffusion advection reaction (dar)